the wings as understudies, ready to step into the shoes of any territory holder who dies on the main stage of group reproduction. We can now see that this may also be their best strategy purely as selfish individuals. As we saw in Chapter 4, we can regard animals as gamblers. The best strategy for a gambler may sometimes be a wait-and-hope strategy, rather than a bull-at-a-gate strategy.

Similarly, the many other examples where animals appear to 'accept' non-reproductive status passively can be explained quite easily by the selfish gene theory. The general form of the explanation is always the same: the individual's best bet is to restrain himself for the moment, in the hope of better chances in the future. A seal who leaves the harem-holders unmolested is not doing it for the good of the group. He is biding his time, waiting for a more propitious moment. Even if the moment never comes and he ends up without descendants, the gamble *might* have paid off, though, with hindsight we can see that for him it did not. And when lemmings flood in their millions away from the centre of a population explosion, they are not doing it in order to reduce the density of the area they leave behind! They are seeking, every selfish one of them, a less crowded place in which to live. The fact that any particular one may fail to find it, and dies, is something we can see with hindsight. It does not alter the likelihood that to stay behind would have been an even worse gamble.

It is a well-documented fact that over-crowding sometimes reduces birth rates. This is sometimes taken to be evidence for Wynne-Edwards's theory. It is nothing of the kind. It is compatible with his theory, and it is also just as compatible with the selfish gene theory. For example, in one experiment mice were put in an outdoor enclosure with plenty of food, and allowed to breed freely. The population grew up to a point, then levelled off. The reason for the levelling-off turned out to be that the females became less fertile as a consequence of over-crowding: they had fewer babies. This kind of effect has often been reported. Its immediate cause is often called 'stress', although giving it a name like that does not of itself help to explain it. In any case, whatever its immediate cause may be, we still have to ask about its ultimate, or evolutionary explanation. Why does natural selection favour females who reduce their birth rate when their population is over-crowded?

Wynne-Edwards's answer is clear. Group selection favours groups in which the females measure the population and adjust their birth rates so